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The picture superiority effect in recognition memory: A developmental study using the response signal procedure

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ABSTRACT

Items studied as pictures are better remembered than items studied as words even when test items are presented as words. The present study examined the development of this picture superiority effect in recognition memory. Four groups ranging in age from 7 to 20 years participated. They studied words and pictures, with test stimuli always presented as words, and time to respond to test stimuli was manipulated. The picture superiority effect showed a clear developmental trend. In the condition in which participants had ample response time, a significant picture superiority effect appeared in all but the youngest group. With short response time, a significant picture superiority effect appeared only among 11- and 20-year-old groups, while a significant reverse of the picture superiority effect was detected in the youngest group. These results were interpreted as suggesting that different memory processes (familiarity and recollection) contribute differently to the picture superiority effect at different stages of development.

1. Introduction

The picture superiority effect in recognition memory tasks refers to the observation that items studied as pictures are better remembered than items studied as words even when targets are presented as words during the testing phase (Mintzer & Snodgrass, 1999; Paivio, 1971). While this robust effect has been extensively studied in adults, very limited knowledge is available about its developmental

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trajectory. Borges, Stepnowsky, and Holt (1977) found that, unlike adults, 4th–6th graders did not show the effect. Unfortunately, however, recognition memory for targets followed their free recall, making it difficult to evaluate the effect of development on the picture superiority effect. The present study is intended to fill this gap. We employed the response signal methodology (Boldini, Russo, Punia, & Avons, 2007) and relied on the theoretical framework provided by dual-process models of memory to guide our predictions.

1.1. A dual-process account of the picture superiority effect in recognition memory

Global matching models (Clark & Gronlund, 1996) invoke a single process to support recognition memory. Recognition decisions are made on the basis of a continuous index of memory strength, familiarity. When a test item exceeds the individual's criterion of memory strength, the stimulus is judged as old; otherwise it is deemed new.

A second class of theories—so-called dual-process models (Diana, Reder, Arndt, & Park, 2006; Yonelinas, 2002)—maintain that a single continuous index such as familiarity is insufficient. Recognition memory can better be accounted for by two processes: the continuous index of strength (familiarity) and a second—recollection, a recall-like process based on recollection of specific qualitative information about the prior occurrence of a target (Jacoby, 1991; Mandler, 1980, 2008). Although differences exist between dual-process models (Mandler, 2008), familiarity is considered to be a rapid, automatic process, sensitive to manipulations of the perceptual features of target items (Lamberts, Brockdorff, & Heit, 2002), while recollection is considered to be a slower, intentional process. Empirical evidence supports the view that recollection is sensitive to manipulations affecting semantic or conceptual encoding, such as level of processing (Boldini, Russo, & Avons, 2004; Ghetti & Angelini, 2008). Furthermore, the use of event-related brain potentials (ERPs) in the study of recognition memory has revealed an early ERP component, at about 300–500 ms, associated with familiarity, and a later component, at about 400–800 ms, associated with recollection (Curran, 2000; Rugg & Curran, 2007).

Boldini et al. (2007) used the response signal procedure to assess the contribution of recollection and familiarity in supporting recognition memory for items studied and tested in either the same or different format (picture–word vs. word–word). In the response signal procedure participants are presented with individual items at test, either studied or new, each followed by a response signal. The response signal indicates the participant must promptly decide whether or not the stimulus is old. Manipulating the interval between test stimulus and response signal allows control of the amount of time available to retrieve the target. Therefore, this technique can be used to plot increments in recognition accuracy as a function of time available to process test stimuli. Because familiarity and recollection processes are presumed to operate at different rates, one critical manipulation is to regulate the time allowed for retrieval prior to decision. On the premise that familiarity is a fast-acting process and recollection a slow-acting process (Mandler, 1980, 2008), it should be possible to trace their contribution to recognition memory by assessing the effect that different variables, expected to selectively influence these two processes, have on recognition accuracy at short and long response times.

Boldini et al. (2007) reasoned that if a perceptually based familiarity process mainly supports recognition memory at short response deadlines, changing the format in which items were presented between study and test should negatively affect recognition accuracy. Consistent with this rationale, a picture 'inferiority' effect was detected at short response-time conditions (studied pictures were less well recognised than studied words), indicating that familiarity primarily supported recognition memory when adults have limited time to process test items. At longer response times, the standard picture superiority effect emerged (studied pictures were better remembered than studied words). The encoding of distinctive sensory/perceptual features of pictures is considered to be at the root of the picture superiority effect (Nelson, 1979). Hence, the distinctive sensory/perceptual features of studied pictures appeared to be more available at long response times. On the basis of this result, Boldini et al. (2007) argued that the "presentation of a target word at test does not necessarily elicit the corresponding studied picture with its distinctive features. Therefore, if it does, this may be regarded as recollection" (p. 122). The picture superiority effect is thus associated with recollection of distinctive features of pictures when test targets appear as words (Dewhurst & Conway, 1994; Schloerscheidt & Rugg, 2004). Overall, Boldini et al. (2007) demonstrated that the picture superiority effect is primarily associated with recollection, while a reverse picture superiority effect primarily indexes the role of familiarity.

1.2. The development of recollection and familiarity processes in recognition memory

The response signal procedure appears to be a useful tool to investigate the development of the picture superiority effect in recognition memory, including assessing the extent to which recollection and familiarity underpin this effect. In principle, at short response times, differences in the magnitude of the reverse picture superiority effect should indicate the extent to which familiarity mostly supports recognition memory at different stages of development. Conversely, the size of the picture superiority effect at long response times should indicate the contribution of recollection.

A few studies have assessed the development of recollection and familiarity processes in recognition memory using methods from studies of young adults (Anooshian, 1999; Billingsley, Smith, & McAndrews, 2002; Brainerd, Holliday, & Reyna, 2004; Ghetti & Angelini, 2008). Anooshian (1999) showed, applying the process dissociation procedure (Jacoby, 1991), that while familiarity was age invariant in a recognition memory task for pictorial information, recollection showed a strong developmental trend between ages 5 and 22. However, this procedure assumes that recollection and familiarity processes independently contribute to performance, an assumption that has been called into question (Curran & Hintzman, 1995; Russo, Cullis, & Parkin, 1998).

Age-related increments in recollection, but not familiarity, have been obtained by Billingsley et al. (2002) using the 'remember' and 'know' procedure (Gardiner, 1988). During the test phase, participants are asked to indicate whether or not each item was presented during the study phase, and if so to indicate any specific information they remember as associated with the item (e.g., any thought elicited by its presentation during learning). 'Remember' vs. 'know' responses are considered to reflect contributions of recollection and familiarity, respectively (but see Rotello, Macmillan, & Reeder, 2004). However, this approach may not be appropriate for children since the instructions for making 'remember' and 'know' responses are complicated, and children may understand and execute them differently than adults (Brainerd et al., 2004).

Brainerd et al. (2004) also attempted to assess the development of vivid recollection and familiarity using a conjoint recognition memory task. Participants are given three types of test items: previously studied items, new items that are semantically associated to studied items, and new items not associated to previously studied items. Participants in different retrieval conditions are asked to accept either previously studied items only, items semantically associated to study items, or both. Estimates of recollection and familiarity are obtained across different retrieval conditions. Results showed that recollection is more influenced by development than familiarity. A possible limitation of this approach is that, as in the case of the 'remember' and 'know' procedure, understanding of the instructions may differ at different ages, thus affecting estimates of recollection and familiarity differently at different ages (Ghetti & Angelini, 2008).

Ghetti and Angelini (2008) used a method proposed by Yonelinas (1994), based on the Response Operating Characteristics curve (ROC). Participants are asked to indicate the degree of confidence in each of their responses (from very confident the item was studied to very confident the item had not been studied). Yonelinas devised a non-linear equation with familiarity and recollection as parameters assumed to contribute independently to recognition memory. This equation is used to generate a theoretical ROC curve; estimates of recollection and familiarity are obtained by fitting the theoretically generated to the empirically obtained ROC curve (but for a criticism of this method see Heathcote, 2003). Using this method, Ghetti and Angelini (2008) showed that recollection increased significantly from age 6 to adulthood, but only when items were incidentally encoded at a semantic level. Familiarity also increased with age, but in contrast to recollection, the asymptote was achieved by age 10.

Overall, despite the limitations indicated, these developmental studies provide a consistent picture: recollection and familiarity appear to have different developmental trajectories. Recollection appears to continue to develop until adulthood, while familiarity is fully operational before adolescence.

1.3. The present study

On the basis of the findings reviewed, it can be predicted that the picture superiority effect should be related to increments in the use of recollection and not familiarity. Using the response signal procedure we intended to assess the development of the picture superiority effect while at the same time exploring the development of the contribution of familiarity and recollection processes to support recognition memory. These contributions are indexed by the modulation of the picture superiority effect as a function of the time available to respond to test stimuli. At short response times, the larger the reverse picture superiority effect, the greater the contribution of familiarity to recognition memory. At long response times, the larger the picture superiority effect, the larger the contribution of recollection.

Unlike Boldini et al. (2007), we had to present verbal stimuli acoustically instead of visually both at study and test because several children during pilot work could not read aloud several of the words that referred to the pictures used. Presenting test stimuli acoustically imposed some constraints on the total amount of time allowed to respond at test. In the short response-time condition this was around 750 ms. At this interval, based on Boldini et al. (2007) results, we did not expect results to show a significant reversal of the picture superiority effect.

Following Boldini et al. (2004, 2007, Exp. 1), we used a go no-go response procedure at test. Participants were asked to respond to each test stimulus only if they thought the stimulus was previously studied. This made the task considerably easier for the younger children, and Boldini et al. (2007) demonstrated that the go no-go and the yes/no techniques yielded comparable results.

2. Method

2.1. Participants

Participants were 27 university students (mean age = 19–6 years; age range 18–2 to 23–3 years), 27 7–year-olds (mean age = 7–3; range 6–8 to 8–0), 27 9–year-olds (mean age = 9–3; range 8–11 to 9–11) and 27 11-year-olds (mean age = 11–3; range 10–10 to 11–11). All children were recruited from in the UK serving a variety of social backgrounds. All were native English speakers. Participants were tested individually. Four 7-year-olds group and one 9-year-old were replaced because they could not follow task instructions.

2.2. Design

A three-factor mixed design was used. Item format at study (picture vs. auditory word) and response time (short: maximum time allowed to respond 750 ms, vs. long: maximum time allowed to respond 2500 ms) were manipulated within subjects (14 participants were tested in the 750 ms condition first, while the order was reversed for the remaining 13). The between-subjects factor was age group.

2.3. Materials

A total of 144 items were selected from Snodgrass and Vanderwart (1980) to be used as target and lure stimuli. A further eight items were selected from the same database as buffer items to be presented at the beginning and at the end of the study lists to reduce the contribution of primacy and recency effects. Two were presented at the beginning of each study list and two at the end. Half of the buffers were presented as pictures and the other half as spoken words. Of the initial 144 items, 48 were presented as pictures (set A) and 48 were presented acoustically as words (set B) in the study phase. The remaining 48 items were presented as lures in the test phase (set C). For each participant, items were randomly allocated to set A, B, and C. For the first study list, 24 items were randomly selected from Set A and 24 items were randomly selected from Set B. The second study list comprised all items not previously presented in the first list. Each study list was created so that an item could only be presented in one format (i.e., an item could not be presented as a picture and a spoken word). At test, two lists were created by selecting the originally 48 studied items plus 24 lures randomly selected from Set C. Items in each set were matched for familiarity and visual complexity (Morrison, Chappell, & Ellis, 1997). For the training phase an additional set of 20 items from Snodgrass and Vanderwart (1980) were selected.

2.4. Procedure

The experiment was conducted in three phases: training phase, study phase and test. Each section began with appropriate instructions on the screen. In addition, for all of the children, the experimenter read out the instructions. Before the main testing session commenced, participants practiced the procedure using a shortened version of the main task. The training phase consisted of the alternate presentation of spoken words and pictures. Stimuli were presented for 500 ms with an inter-stimulus interval of 1 s. Participants were asked to memorize as many items as possible for a later unspecified memory test. To ensure that all items were being attended to, participants were requested to verbally name all of the items studied (i.e. both pictures and acoustically presented words). At test, the 10 studied items along with 10 lures were presented in a random order. Each item was presented as a spoken word through headphones and the duration of each word was no longer than 250 ms. After the appropriate time lag from the onset of a test stimulus (i.e., either 250 or 2000 ms) a row of asterisks appeared in the centre of the computer screen for 500 ms and participants were asked to decide whether the item had been previously studied. They were told to press the spacebar if they had previously studied the item, or to do nothing otherwise. The time allowed for responding to each test item remained constant at 500 ms regardless of whether the subject was being tested in the fast or slow condition (thus the maximum allowed time to respond were 750 and 2500 ms in the short and long response deadline conditions, respectively). When the time allowed for responding was over, the row of asterisks disappeared from view and participants heard a 'beep' alert. This notified participants that the time for responding had elapsed. Between consecutive items there was a pause of 1 s. All pictures were presented in the centre of the computer screen and all the words were presented through a set of headphones. All words were recorded in an English, female voice.

The main task followed the training phase. The procedure was the same as that used in the training phase with the exception that participants were presented with two buffer items, followed by the 48 target items, then two buffer items. As in the training phase, participants were asked to memorize as many items as possible for a later memory test. Participants were given a 1-min task-filled delay before commencing the test phase. At test, the 48 original items were randomly presented with 24 distractor items. All stimuli were presented as spoken words and in a random order that varied across participants. The procedure was then repeated for the alternate response-time condition. On completion of both study/test conditions, participants were debriefed and thanked for their participation. Overall the procedure lasted approximately 1 h, with children tested in two 30-min sessions.

3. Results

Mean proportions of hits, false alarms and A′ scores per groups and conditions are presented in Table 1. A′ scores provide an index of recognition memory accuracy independent of response bias. A score of 1 indicates perfect accuracy, while a score of 0.5 indicates chance performance. A 2 (stimulus type: words vs. pictures; within-subjects) × 2 (response time: short vs. long; within-subjects) × 4 (age group; between-subjects) mixed analysis of variance (ANOVA) was carried out on A′ scores (Donaldson, 1992).

There was a significant main effect of age, $F(3, 104) = 25.56$, $MSE = 0.025$, $p < .01$, indicating that accuracy increased with age. The main effect of response time was significant, $F(1, 104) = 86.80$, $MSE = 0.01$, $p < .01$, indicating that accuracy was higher at the long response time. The main effect of stimulus type was also significant, $F(1, 104) = 33.93$, $MSE = 0.004$, $p < .01$, indicating that performance was better for studied pictures than words.

Several interactions were also significant. The interaction between age and response time was significant, $F(3, 104) = 4.71$, $MSE = 0.01$, $p < .01$, indicating that the effect of response time on accuracy differed as a function of age. Simple comparisons between short and long response times showed that accuracy did not reach significance among adults, $t(26) = 1.96$, $p = .064$, while for all remaining

Table 1
(1a) Proportion of hits, false alarms and (1b) A′ scores for items presented at study as pictures and as words, by age and response-time condition.

	Fast			Slow		
	Hits pictures	Hits words	FA	Hits pictures	Hits words	FA
(1a)						
Adults	.776 (.14)	.568 (.18)	.125 (.10)	.850 (.13)	.619 (.17)	.105 (.11)
11-year	.647 (.14)	.576 (.16)	.301 (.17)	.802 (.17)	.620 (.17)	.161 (.10)
9-year	.571 (.22)	.546 (.20)	.289 (.20)	.744 (.16)	.627 (.18)	.222 (.18)
7-year	.414 (.25)	.466 (.23)	.318 (.24)	.588 (.24)	.574 (.21)	.267 (.18)

	Fast		Slow	
	A′ pictures	A′ words	A′ pictures	A′ words
(1b)				
Adults	.896 (.05)	.818 (.08)	.926 (.06)	.846 (.09)
11-year	.749 (.11)	.713 (.11)	.887 (.08)	.818 (.09)
9-year	.720 (.13)	.709 (.12)	.836 (.10)	.790 (.11)
7-year	.608 (.12)	.649 (.12)	.746 (.13)	.737 (.11)

Note. Standard deviations appear in parentheses.

groups accuracy was significantly higher at the long than at the short times, $ts(26) > 4.26$, $p < .01$. The interaction between stimulus type and response deadline was significant, $F(1, 104) = 8.03$, $MSE = 0.003$, $p < .01$, indicating that the advantage of pictures over words was larger in the long response-time condition. Similarly, the interaction between stimulus type and age was significant, $F(3, 104) = 10.62$, $MSE = 0.004$, $p < .01$, indicating that the advantage of pictures over words increased as a function of age. The three-way interaction was not significant, $F < 1$.

To provide a clearer picture of the effect of age and response time on the size of the picture superiority effect, an age by response time mixed ANOVA was carried out on the difference of A′ scores between items studied as pictures and those studied as words (see Fig. 1). This further analysis was performed since difference scores are more sensitive measures for the picture superiority effect independently of overall level of performance. The effect of age was significant, $F(3, 104) = 10.91$, $MSE = 0.008$, $p < .01$. Analyses within age groups revealed a significant picture superiority effect among adults (the 95% confidence interval (CI) ranged from 0.061 to 0.096), 11-year-olds (the 95% CI ranged from 0.033 to 0.071) and 9-year-olds (the 95% CI ranged from 0.007 to 0.051). However, the effect was not significant

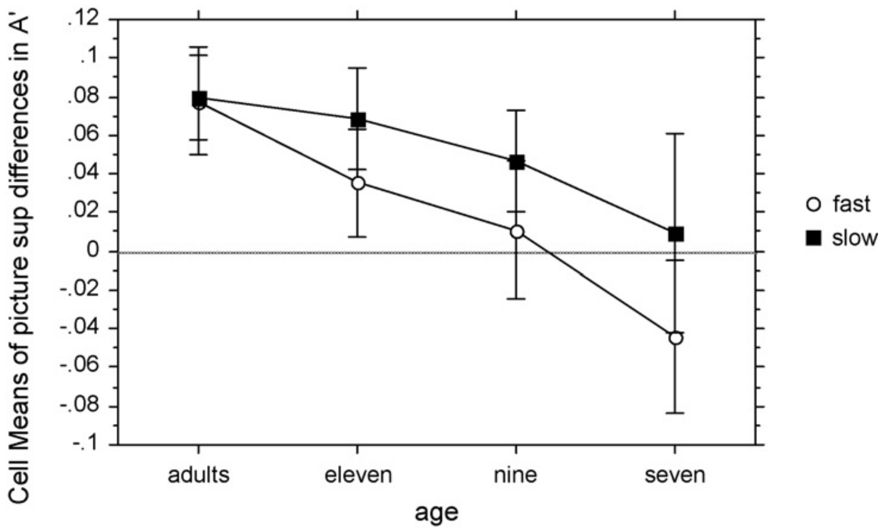

Fig. 1. Plot of the difference between A′ scores for items studied as pictures and as words, as a function of slow/fast response-time condition and age. Positive scores indicate that recognition memory accuracy was greater for pictures than for words. Bars represent 95% confidence intervals.

among 7-year-olds (the 95% CI ranged from −0.050 to 0.015). The main effect of response time was significant, $F(1, 104) = 8.44$, $MSE = 0.006$, $p < .01$, with the picture superiority effect larger at the long response time (the 95% CI ranged from 0.034 to 0.067) than the short time (the 95% CI ranged from 0.002 to 0.038).

Although the age by response time interaction was not significant, $F(3, 104) = 1.02$, $MSE = 0.006$, $p > .10$, it is relevant, as explained earlier, to further assess differences in the picture superiority effect within age groups at each response-time condition.

At the short response time, there was a significant reduction in the reverse picture superiority effect as a function of age, $F(1, 104) = 29.18$, $MSE = 0.007$, $p < .01$, for the linear trend. A significant reverse of the picture superiority effect was detected among 7-year-olds (the 95% CI ranged from −0.084 to −0.004). Among 9-year-olds the reverse picture superiority effect was not significant (the 95% CI ranged from −0.025 to 0.047), while among 11-year-olds and adults there was a significant picture superiority effect (the 95% CI ranged from 0.008 to 0.064 and from 0.050 to 0.106, respectively).

At the long response time, there was a significant increase in the picture superiority effect as a function of age, $F(1, 104) = 10.02$, $MSE = 0.007$, $p < .01$ for the linear trend. The picture superiority effect was not significant among 7-year-olds (the 95% CI ranged from −0.042 to 0.061) but was significant among 9-year-olds (the 95% CI ranged from 0.020 to 0.073), 11-year-olds (the 95% CI ranged from 0.042 to 0.095) and adults (the 95% CI ranged from 0.058 to 0.101).

4. Discussion

The results indicate a developmental trend in recognition accuracy as well as in the picture superiority effect. The latter effect was not significant among 7-year-old children, but it was significant among older age groups with the largest effect being detected among young adults. Although other studies have investigated the picture superiority effect in recall tasks (Whitehouse, Maybery, & Durkin, 2006), the present study provides the first clear demonstration of a developmental trend in the picture superiority effect in recognition memory. Furthermore, the size of the picture superiority effect can be considered as an indicator of the role that recollection plays in supporting recognition memory performance. Therefore, in line with previous studies (Anooshian, 1999; Billingsley et al., 2002; Brainerd et al., 2004; Ghetti & Angelini, 2008), the present study indicates an age-related increase in the role of recollection in supporting recognition memory.

This conclusion can be further qualified by examining the outcome of the signal response procedure in more detail. At the short response time, a significant reverse picture superiority effect emerged only among 7-year-old children. This effect was not significant among 9-year-old children. On the other hand, a significant picture superiority effect was detected among 11-year-old children and adults. As explained earlier, the presence of a reverse picture superiority effect at the short response time is an indication that familiarity is predominantly supporting recognition memory. Hence, from the present results it can be inferred that, when participants had minimal amount of time to retrieve the target information at test, recognition memory decisions in the youngest age group, unlike the other age groups, were primarily based on familiarity.

At the long response time, the picture superiority effect was significant in all but the 7-year-old group. If the larger the picture superiority effect the stronger the contribution of recollection in supporting recognition memory performance, it can be inferred that, when people have ample time to retrieve information at test, recollection shows a strong developmental trend, with minimal contribution to recognition memory performance in the youngest group.

Summarising, using the response signal procedure to examine the development of the picture superiority effect, recollection showed a clear developmental trend. However, based on the results obtained in the short response-time condition, under relatively severe time constraints to retrieve target items, 7-year-old children, unlike the other age groups, appeared to base recognition memory primarily on familiarity. Overall, the present findings are mainly consistent with those obtained using different methodologies in indicating that recollection shows a clear developmental trend in supporting recognition memory.

With regard to the lack of a reverse picture superiority effect in all age groups but the youngest one at the short response time, this may appear at variance with the results obtained by Boldini et al.

(2007). In that study, adults repeatedly showed a reliable reverse picture superiority effect at short response times. However, Boldini et al. (2007) presented verbal stimuli visually at test. This reduced the amount of time to respond to a test word to about 550/600 ms, thus reducing the opportunity to recollect features of target items to support recognition performance. In the present experiment, with acoustic presentation of test stimuli, the total amount of time available to respond to a test word was about 750 ms. Hence, with this extra time to respond it was more likely that at the short response times, recollection supported recognition decisions, at least in the older groups. This hypothesis is supported by the observation that in the adult group the size of the picture superiority effect was almost identical in the fast and slow response-time conditions. This methodological difference between Boldini et al. (2007) and the present study most likely constrained the potential effect that the short response-time condition could have had on the size of the reverse picture superiority effect. Hence, with more severe time constraints to respond at short response times, it may be possible to obtain a clearer picture of the extent to which the reverse picture superiority effect, as an index of familiarity supporting recognition memory, is affected by development. In the present study we had to use acoustic presentation of test stimuli, for the reasons explained earlier. Thus we had to use less than optimal testing conditions, at least in the short response-time condition.

In conclusion, in seeking to understand the development of the picture superiority effect using the response signal methodology, we showed converging evidence consistent with results obtained using other paradigms that should make it possible to quantify the relative contribution of familiarity and recollection (Anooshian, 1999; Brainerd et al., 2004; Ghetti & Angelini, 2008) in recognition memory. Based on the present results we can conclude that recollection showed a clear developmental trend and that familiarity had a stronger role in supporting recognition memory among young children.

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